

JNTU Hyderabad

MID-TERM EXAMINATION ? 2026

Computer Networks ? Mid-Term Examination, 2026

Subject Code: CS501 Semester: 5 Regulation: R20

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Network Models and Physical Layer)

1. Compare the OSI and TCP/IP reference models with a neat layered diagram. [12 Marks]
2. How is Network Models and Physical Layer implemented? Give an algorithm / procedure and explain. [8 Marks]
3. Discuss the engineering applications of Network Models and Physical Layer. [12 Marks]

SECTION B (Unit 2: Data Link Layer)

4. Explain the Stop-and-Wait and Sliding Window flow control protocols with examples. [12 Marks]
5. What is CSMA/CD? Explain how Ethernet handles collisions. [12 Marks]
6. Describe the working principle of Data Link Layer with necessary equations. [12 Marks]

SECTION C (Unit 3: Network Layer and Routing)

7. Explain Dijkstra's and Bellman-Ford routing algorithms with examples. [12 Marks]
8. Derive the expression related to Network Layer and Routing and explain each term. [12 Marks]
9. List the advantages and limitations of Network Layer and Routing. Explain any two in detail. [12 Marks]

SECTION D (Unit 4: Transport Layer)

10. Compare TCP and UDP transport protocols. Explain TCP connection establishment and termination. [12 Marks]
11. Describe the working principle of Transport Layer with necessary equations. [12 Marks]
12. How is Transport Layer implemented? Give an algorithm / procedure and explain. [12 Marks]

SECTION E (Unit 5: Application Layer and Network Security)

13. Explain the DNS, HTTP and SMTP application layer protocols with their message formats. [12 Marks]
14. What is a firewall? Explain the types of firewalls and common network security

attacks. [12 Marks]

15. Write short notes on Application Layer and Network Security. [8 Marks]

*** End of Question Paper ***